

IE5203 Decision Analysis Assignment 2 Solutions

Question 1.

As Linda's wealth utility function =

$$u(w) = \begin{cases} \ln(1+w) & w \geq 0 \\ w & w < 0. \end{cases}$$

$$u'(w) = \begin{cases} \frac{1}{1+w} & w \geq 0 \\ 1 & w < 0 \end{cases} \quad u''(w) = \begin{cases} \frac{-1}{(1+w)^2} & w \geq 0 \\ 0 & w < 0. \end{cases}$$

Degree of absolute risk aversion =

$$r(w) = \frac{-u''(w)}{u'(w)} = \begin{cases} \frac{1}{1+w} & w \geq 0 \\ 0 & w < 0. \end{cases}$$

Hence, at current wealth $w = \$500$, Linda's degree of risk absolute aversion = $1/501 \text{ } \$^{-1}$.

The unit for degree of absolute risk aversion is $\$^{-1}$.

Question 2.

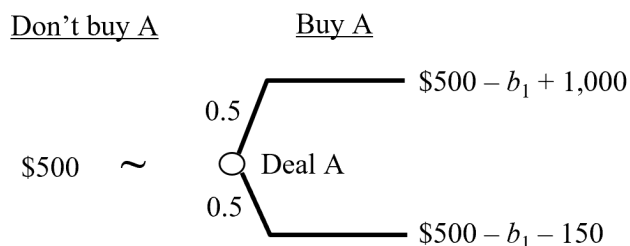
Linda is risk averse at her current wealth, as her degree of absolute risk aversion is positive.

Alternatively, Linda is risk averse for all w , as $u(w)$ is an increasing concave function for all w .

Question 3.

Linda's current wealth = \$500.

Let her personal indifferent buying price for Deal A alone = b_1 .



When Linda is indifferent between buying and not buying Deal A:

$$u(500) = 0.5 u(500 - b_1 + 1000) + 0.5 u(500 - b_1 - 150),$$

$$u(500) = 0.5 u(1500 - b_1) + 0.5 u(350 - b_1).$$

Assume $b_1 \leq 350$:

$$\ln(501) = 0.5(\ln(1501 - b_1) + \ln(351 - b_1)),$$

$$501^2 = (1501 - b_1)(351 - b_1).$$

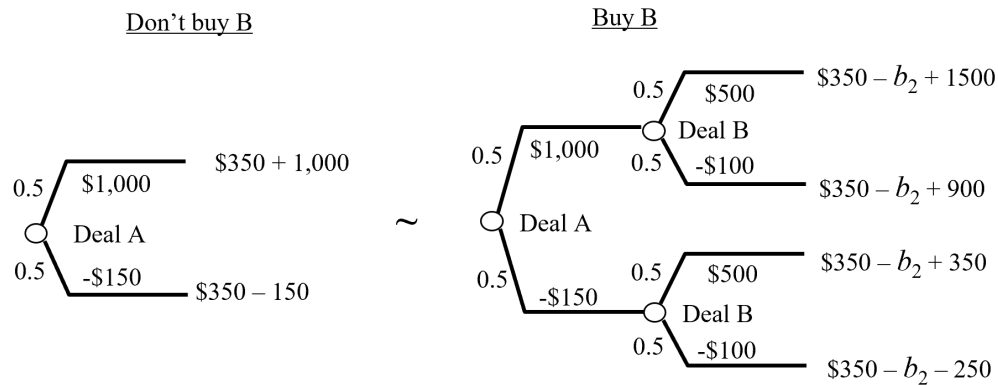
Solving the quadratic equation with a solver: $b_1 = 163.36$.

Linda's current personal indifferent buying price for Deal A is **\$ 163.36**.

Question 4.

Linda now owns Deal A and \$350 in cash.

Let her personal indifferent buying price for Deal B alone = b_2 .



When Linda is indifferent between buying and not buying Deal B:

$$\begin{aligned}
 0.5 u(350 + 1000) + 0.5 u(350 - 150) &= 0.5 (0.5 u(350 - b_2 + 1500) + 0.5 u(350 - b_2 + 900)) \\
 &\quad + 0.5 (0.5 u(350 - b_2 + 350) + 0.5 u(350 - b_2 - 250)), \\
 u(1350) + u(200) &= 0.5 (u(1850 - b_2) + u(1250 - b_2)) \\
 &\quad + u(700 - b_2) + u(100 - b_2)).
 \end{aligned}$$

Assume $b_2 \leq 100$:

$$\begin{aligned}
 \ln(1351) + \ln(201) &= 0.5 (\ln(1851 - b_2) + \ln(1251 - b_2) \\
 &\quad + \ln(701 - b_2) + \ln(101 - b_2)), \\
 (1351)^2 (201)^2 &= (1851 - b_2)(1251 - b_2)(701 - b_2)(101 - b_2).
 \end{aligned}$$

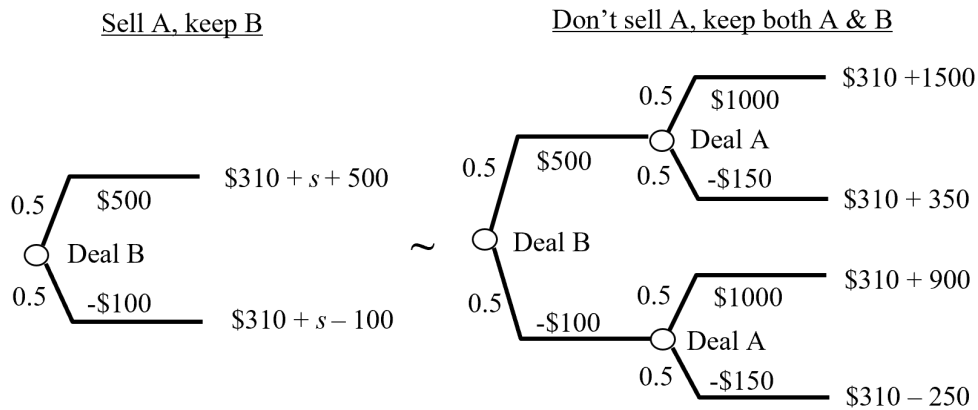
Solving the quartic equation with a solver: $b_2 = 48.81$.

After buying Deal A for \$150, Linda's personal indifferent buying price for Deal B is **\$ 48.81**.

Question 5.

Linda now owns Deal A, Deal B, and \$310 in cash.

Let her personal indifferent selling price for just Deal B = s .



When Linda is indifferent between selling and not selling Deal A:

$$\begin{aligned}
 0.5u(310 + s + 500) + 0.5u(310 + s - 100) &= 0.5(0.5u(310 + 1500) + 0.5u(310 + 350)) \\
 &\quad + 0.5(0.5u(310 + 900) + 0.5u(310 - 250)), \\
 u(810 + s) + u(210 + s) &= 0.5u(1810) + 0.5u(660) + 0.5u(1210) + 0.5u(60).
 \end{aligned}$$

Assume $s \geq -210$:

$$\begin{aligned}
 \ln(811 + s) + \ln(211 + s) &= 0.5 \ln(1811) + 0.5 \ln(661) + 0.5 \ln(1211) + 0.5 \ln(61), \\
 (811 + s)^2(211 + s)^2 &= (1811)(661)(1211)(61).
 \end{aligned}$$

Solving the quartic equation with a solver: $s = 111.39$.

Linda's personal indifferent selling price for Deal A alone is **\$ 111.39**.